



Unpacking the Climate Hiatus: Where Did the Global Warming Heat Go?

Abstract

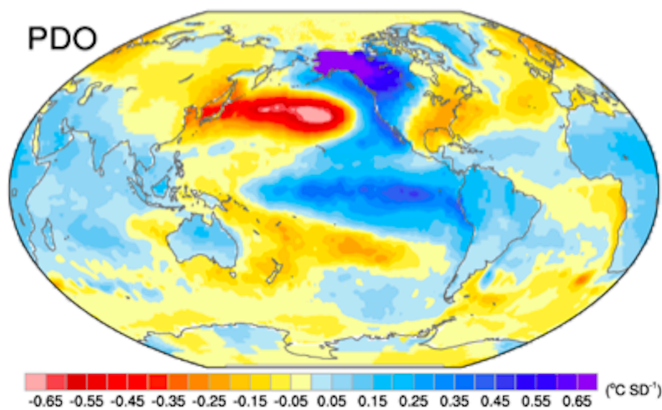
Between 2009 and 2014, several harsh winters in Europe and the eastern United States led people to ask, "Where is global warming?" Climate scientist Kevin Trenberth explained that these cold spots covered only about 2% of the globe, while the Arctic was unusually warm. His research traced the pattern to the tropical Pacific, where changes in the ocean and atmosphere sent waves through the jet stream and carried heat deep into the ocean.

Introduction

Weather is never normal. It swings between cooler and warmer periods as weather systems move around the globe. The northern winter of 2009-10 was exceptionally cold in Europe, yet it was one of the warmest winters globally. In 2013-14, the eastern United States shivered through a polar vortex while Alaska and the West Coast set heat records and California fell into a record-breaking drought. Global mean temperatures had climbed steadily from the 1970s until 1998 but then seemed to stall. Some people used this so-called hiatus to claim global warming had stopped. Trenberth and colleagues set out to explain what was really happening.

How Do Scientists Study This?

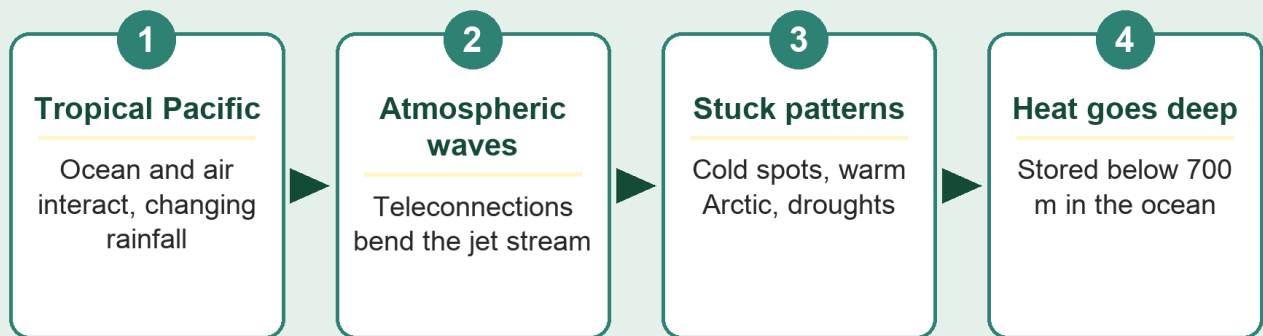
Trenberth's team built on theory developed in the 1980s about how the tropical Pacific Ocean influences weather worldwide. During El Niño, interactions between ocean and atmosphere warm the central and eastern tropical Pacific over a year to 18 months. This shifts where tropical storms and thunderstorms form, which creates large-scale waves in the atmosphere that bend the jet stream and storm tracks, sometimes across the Americas into the Atlantic. Scientists call these long-distance links teleconnections. The team applied this idea to the hiatus, comparing global temperature records, Arctic temperatures, Antarctic sea ice, and ocean heat at different depths, and examined the role of the Pacific Decadal Oscillation.



FROM THE ARTICLE

The Pacific Decadal Oscillation pattern of sea surface temperatures. Image: Kevin Trenberth, author provided.

How the Pacific shaped the hiatus



Study the Process

Start in the tropics

Scientists focused on the tropical Pacific, a key source of changes in rainfall and atmospheric waves.

Follow the waves

They traced how teleconnections bend the jet stream, creating cold spots and warm spots far away.

Track the heat

They measured where heat went, including the deep ocean below 700 meters.

What Did They Discover?

The team found that global warming continued but showed up in different ways. When large atmospheric waves get stuck, the same weather repeats: cold outbreaks keep hitting one region while drought or floods persist elsewhere. Cold European winters went along with a warm Arctic, and the Arctic likely warmed more than expected because of these wave patterns. So Arctic amplification was a result of the waves, not their cause. The same patterns explain Antarctic sea ice shrinking near the Antarctic Peninsula but expanding in the South Pacific. Meanwhile, changes in winds pushed heat deeper than 700 meters in the ocean. The eastern Pacific was the largest region that did not warm in the 2000s.

The hiatus by the numbers

~2%

of the globe covered by the exceptionally cold spots

1998

year after which surface warming appeared to stall

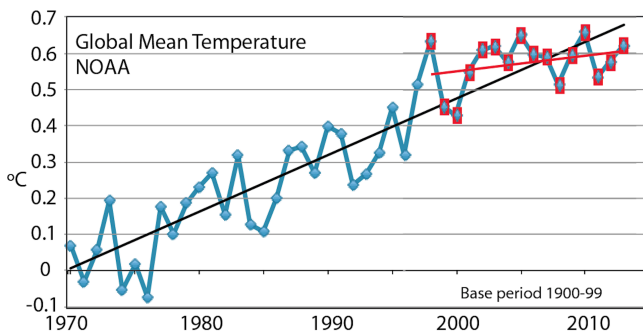
700 m

depth below which extra heat was stored

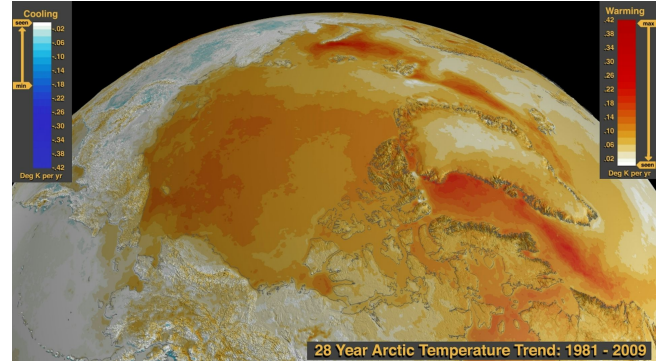
The Pacific Decadal Oscillation, a natural pattern that has operated for centuries, may itself be altered by climate change. Predicting it could improve warnings about extremes.

Look at the Evidence

FROM THE ARTICLE



Global mean surface temperature relative to 1900-99, with trends for 1970-2013 and 1998-2013. From Trenberth et al. (2014), Nature Climate Change.



Arctic temperature trend, 1981-2009: a warmer Arctic accompanied colder Northern Hemisphere winters. Image: NASA.

Takeaways

Local cold, global warmth

Record cold spots covered about 2% of Earth while the Arctic stayed unusually warm.

The Pacific connects

Changes in the tropical Pacific send waves through the atmosphere that affect weather worldwide.

Heat went deep

The planet kept warming, but heat was stored below 700 meters in the ocean.

Key Terms

Jet stream

A fast, narrow band of wind high in the atmosphere that steers weather systems.

Teleconnection

A link between weather or climate in one region and conditions in a distant region.

Polar vortex

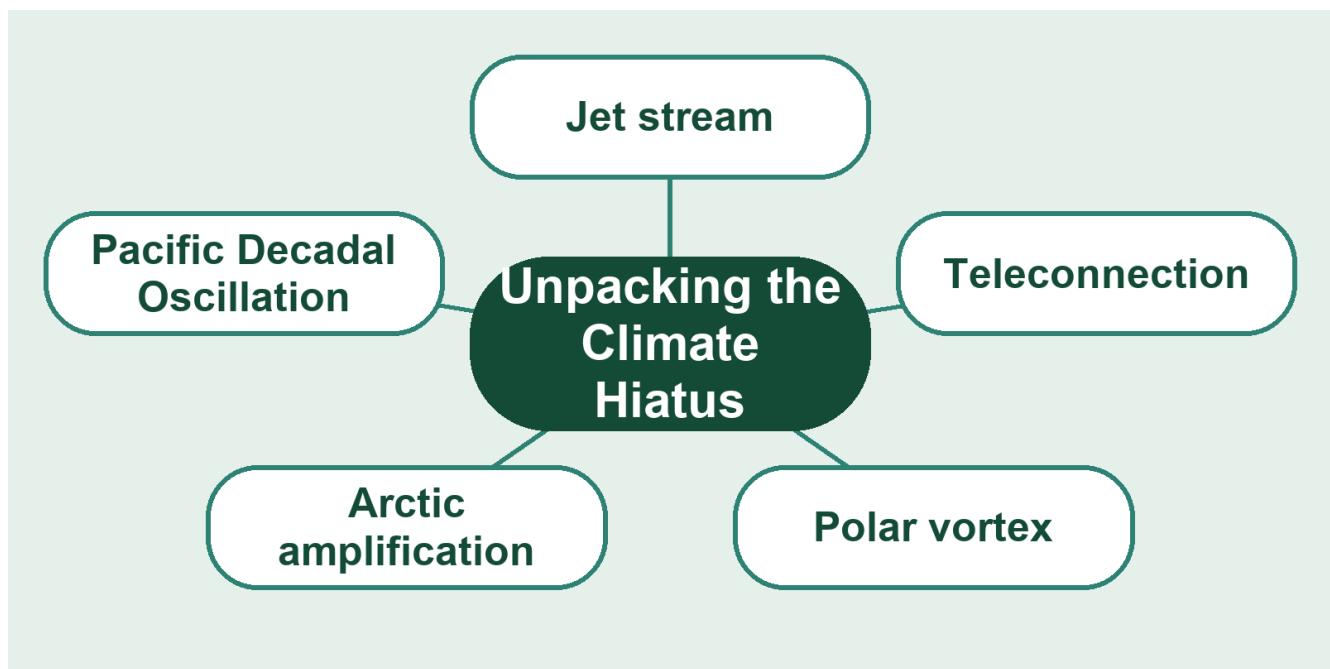
A large area of cold air around the poles that can spill southward and bring extreme cold.

Arctic amplification

The tendency of the Arctic to warm faster than the rest of the planet.

Pacific Decadal Oscillation

A natural, long-lasting pattern of ocean temperature change in the Pacific that affects climate for years to decades.



Activity

Weather Wave Tracker

Students model how a stuck wave in the jet stream creates cold spots and warm spots at the same time.

Materials: Long rope or jump rope; World map; Sticky notes labeled cold, warm, dry, and wet

1. Two students hold a rope and create a wave pattern to represent the jet stream across North America.
2. Place the map beneath the rope. Areas where the rope dips south get cold notes; areas where it bulges north get warm notes.
3. Freeze the wave in place. Explain what happens to weather when the pattern stays stuck for weeks.
4. Compare with the 2013-14 winter: a cold East Coast and a warm, dry West Coast.
5. Write a response to someone who says a cold winter proves global warming is not happening.

References

Trenberth, K. (2014). What caused the 'pause' in global warming? The Conversation.
<https://doi.org/10.64628/AAI.9pkwx3tf>

Trenberth, K. E., et al. (2014). Nature Climate Change, as cited in the article above.